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EE1060: Differential Equations and Transform Techniques

Lecture 27: Impulse and Impulse Response

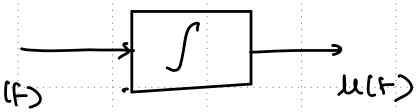
Topics :

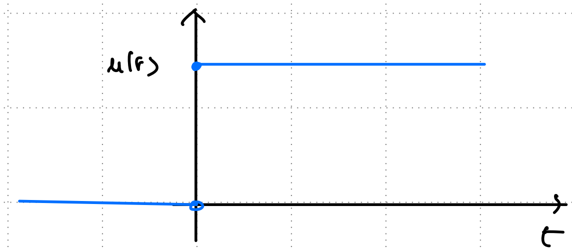
1. Introduction to Impulse signal
2. Impulse Response

Introduction to Impulse Signal

for System :- Basic building block is '$\int$'

$\delta(t) \leftarrow \underline{x(t)}$





Version-03 :-

generalised function

$g(x) \leftarrow$ "

$\underline{\phi(x)} \leftarrow$ test $\underline{\text{function}}$.

$$\delta(t) : g(x) : \int_{-\infty}^{\infty} g(x)\,\underline{\phi}(x)\,dx = \phi(0)$$

Version-02:-

$$\int_{-\infty}^{t} \delta(t) \cdot dt = 1 \quad \forall\, t > 0$$
$$= u(t)$$

Version-01:-

$x(t) = \delta(t)$

a) '0' for $t < 0$

b) '0' for $t > 0$.

$\underline{\text{c)}}$ $\lim_{\epsilon \to 0} \int_{-\epsilon}^{\epsilon} \delta(t)\,dt = 1$

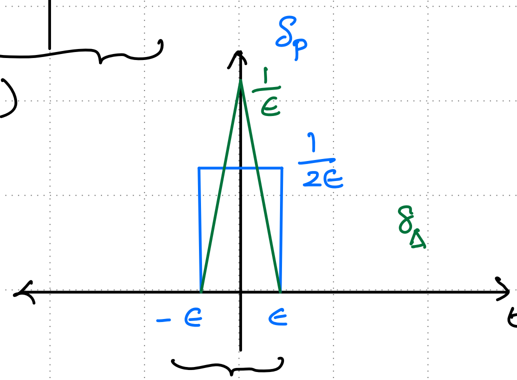
$= u(\epsilon) - u(-\epsilon)$

$= 1 - 0 = \underline{1}$

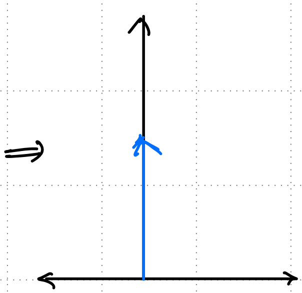
$\underline{\phi(x)} \longrightarrow$ a) Smooth f/n's (differentiable up to any order)

b) finite support

def :- $$\int_{-\infty}^{\infty} \delta(t-\tau)\, f(t)\, dt = f(\tau)$$



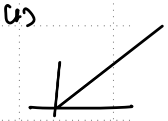
$\Longrightarrow$

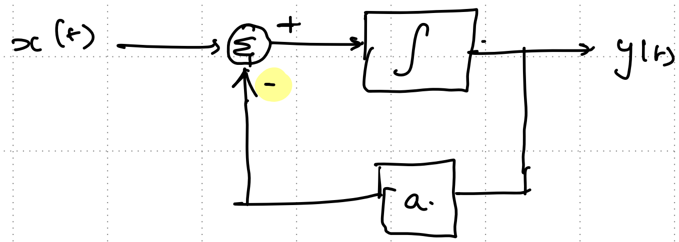


Impulse Response $h(t)$

$h(t) \longrightarrow$ impulse response of the system $\underline{\underline{\delta(t)}}$

$\delta(t) \longrightarrow \int \longrightarrow u(t)$

$u(t) \longrightarrow \int \longrightarrow$ 

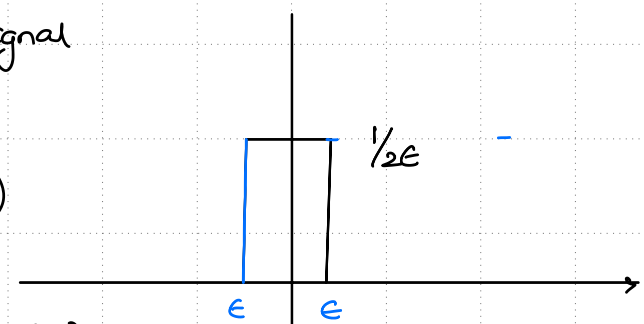


$$\frac{dy}{dt} + ay = x(t)$$

$$\frac{dy}{dt} + ay = \underline{\underline{\delta(t)}}$$

Step response :- ? $\frac{1}{a}(1-e^{-at})$

$\underline{h(t)}$:- limiting sense of a pulse signal



$$\delta_p(t) = \left(\frac{1}{a}\right)\frac{1}{2\epsilon}\Big(u(t+\epsilon) - u(t-\epsilon)\Big)$$

$$h_P(t) = \frac{1}{2\epsilon a}\left(1 - e^{-a(t+\epsilon)} - \left\{1 - e^{-a(t-\epsilon)}\right\}\right)$$

$$= \frac{e^{-at}\left(e^{a\epsilon} - e^{-a\epsilon}\right)}{2\epsilon a}$$

$$h(t) = \lim_{\epsilon \to 0} h_p(t) = \underline{\underline{e^{-at}}}$$

Impulse Response

Stability analysis:-

a) $h(t) \longrightarrow 0$ as $t \longrightarrow \infty$, System is stable

b) $h(t)$ is Constant $(\neq 0)$ as $t \longrightarrow \infty$, System is marginally stable.

c) $h(t) \longrightarrow \infty$ as $t \longrightarrow \infty$, System is unstable.

for a first order system $\dfrac{dy}{dt} + ay = \delta(t)$

Stable when $a > 0$

marginally stable when $a = 0$

unstable when $a < 0$.

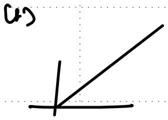
(when $a = 0$, the system is an integrator ;

when input to integrator is step

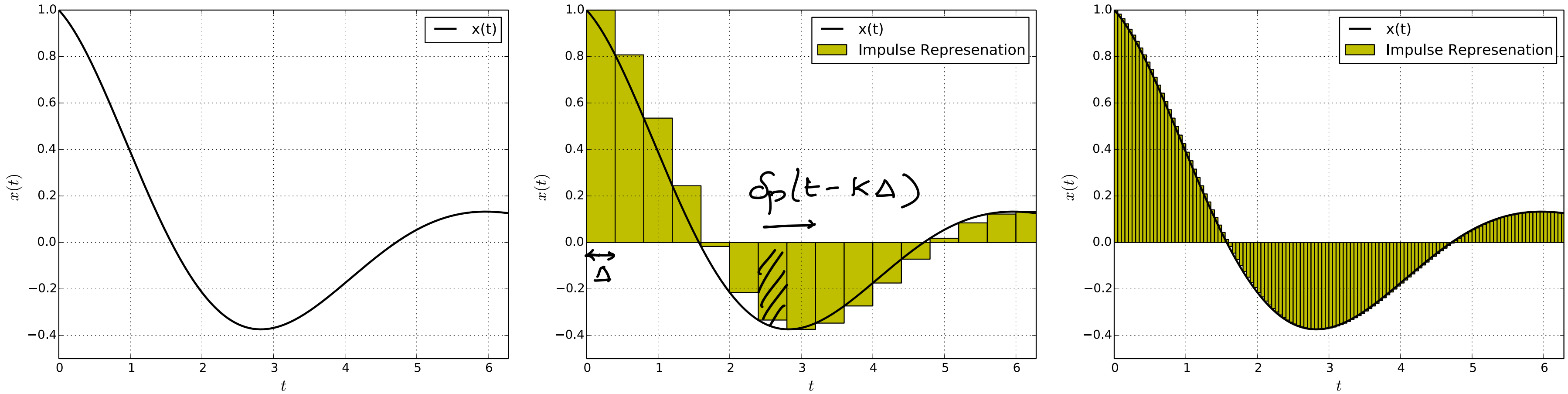
o/p is ramp).

$\delta(t) \longrightarrow \int \longrightarrow u(t)$

$u(t) \longrightarrow \int \longrightarrow$



Representation of Input Signal using Impulses



(a) Original signal $x(t)$

(b) Representation of $x(t)$ using impulses of width 0.40

(c) Representation of $x(t)$ using impulses of width 0.05

Fig. 1: Representation of $x(t)$ using rectangular impulse

The input signal $x(t)$ can be approximated using the pulse signal as

$$\approx \sum_{k=-\infty}^{\infty} x(k\Delta)\, \delta_p(t - k\Delta)\, \Delta$$

as the width of the pulse reduces.

$$\text{as } \Delta \to 0, \quad x(t) = \int_{-\infty}^{\infty} x(\tau)\, \delta(t-\tau)\, d\tau.$$